

Empowering Mobile Networks Security Resilience by using Post-Quantum Cryptography

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Abstract—The transition to a cloud-native 5G Service-Based Architecture (SBA) improves scalability but exposes control-plane signaling to emerging quantum threats, including Harvest-Now, Decrypt-Later (HN DL) attacks. While NIST has standardized post-quantum cryptography (PQC), practical, deployable integration in operational 5G cores remains underexplored. This work experimentally integrates NIST-standardized ML-KEM-768 and ML-DSA into an open-source 5G core (free5GC) using a sidecar proxy pattern, preserving unmodified network functions (NFs). Implemented on free5GC, we compare three deployments: (i) native HTTPS/TLS, (ii) TLS sidecar, and (iii) PQC-enabled sidecar. Measurements at the HTTP/2 request-response boundary over repeated independent runs show that PQC increases end-to-end Service-Based Interface (SBI) latency to ~ 54 ms, adding a deterministic 48–49 ms overhead relative to the classical baseline, while maintaining tightly bounded variance ($IQR \leq 0.2$ ms, $CV < 0.4\%$). We also quantify the impact of Certification Authority (CA) security levels, identifying certificate validation as a tunable contributor to overall delay. Overall, the results demonstrate that sidecar-based PQC insertion enables a non-disruptive and operationally predictable migration path for quantum-resilient 5G signaling.

I. INTRODUCTION

The deployment of 5th Generation Mobile Network (5G) networks represents a paradigm shift in telecommunications, supporting a massive ecosystem of billions of devices. Central to this evolution is the Service-Based Architecture (SBA), which replaced traditional point-to-point interfaces with a cloud-native, microservices-oriented approach. While this transition to Hypertext Transfer Protocol (HTTP)/2 and Transport Layer Security (TLS) enhances scalability, it also centralizes sensitive cryptographic operations at critical core network functions Network Functions (NFs) [1], [2].

As research progresses toward 6th Generation Mobile Network (6G) networks, the security foundations of these infrastructures face an existential threat: the advancement of quantum computing. Current 5G security relies heavily on public-key cryptography, specifically Rivest–Shamir–Adleman (RSA) and Elliptic-curve cryptography (ECC), which are vulnerable to Cryptographically Relevant Quantum Computers (CRQCs). Attacks such as Harvest Now, Decrypt Later (HN DL) [3] underscore the urgency, as adversaries may already be capturing sensitive 5G signaling to decrypt once powerful quantum machines become available. This poses a long-term risk to subscriber identities and authentication credentials stored in Unified Data Management (UDM) modules [4], [2].

To mitigate these risks, the National Institute of Standards and Technology (NIST) finalized the first Post-Quantum Cryptography Post-Quantum Cryptography (PQC) standards in 2024, emphasizing Key Encapsulation Mechanism (KEM) and Digital Signature Algorithms (DSA). However, a significant gap remains between theoretical standardization and practical implementation in high-throughput environments like the 5G Core [5], [6].

This work addresses this gap by presenting a comprehensive evaluation of PQC integration into a cloud-native 5G Core. We propose a sidecar proxy architecture that enables quantum-safe communication between NFs without modifying the legacy codebase. Specifically, we integrate ML-KEM[7] and ML-DSA[8] into the free5GC platform. Our main contribution is a rigorous experimental comparison across three scenarios: (i) a baseline Native5G Core; (ii) a TLS-based sidecar; and (iii) our PQC-enabled sidecar architecture. We provide detailed insights into the impact of larger PQC keys on control plane latency, handshake performance, and resource consumption (Central Processing Unit (CPU) /Memory), offering a roadmap for future quantum-resistant telecommunications infrastructure.

To the best of our knowledge, this work is among the first fully experimental integrations of NIST-standardized PQC algorithms into a cloud-native 5G Core, using a sidecar proxy pattern that preserves unmodified NF implementations. Unlike prior studies that focus on theoretical security analysis, protocol-level proposals, or hybrid cryptographic simulations, our approach provides a deployable migration strategy within a production-grade free5GC environment.

Beyond reporting end-to-end latency, we experimentally isolate cryptographic processing overhead from wrapper-induced proxy costs, quantify deterministic control-plane behavior under PQC, and evaluate the impact of different Certificate Authority (CA) security parameter levels. This enables a practical characterization of performance–security trade-offs and demonstrates crypto-agility in SBA signaling without intrusive architectural changes.

The remainder of this paper is organized as follows. Section II discusses related work on PQC and its adoption in 5G and cloud-native deployments. Section III presents the threat model and security goals. Section IV describes the proposed sidecar-based architecture for protecting SBI and extending coverage to N2 signaling. Section V details the implementation to empower Free5GC with PQC. Section VI

presents the experimental methodology, including evaluation scenarios and measurement approach. Section VII reports and analyzes the results, and Section VIII concludes the paper and outlines future work.

II. RELATED WORK

This section reviews recent literature on securing 5G and 6G networks against emerging quantum computing threats. The discussed works highlight key areas, including the security analysis of Service-Based Architectures SBA, the adoption of Post-Quantum Cryptography PQC in telecommunications, and the practical vulnerability and performance assessments of open-source 5G core deployments.

Mahyoub et al. [9] present a security analysis of critical interfaces in the Fifth Generation architecture, identifying interfaces that exchange sensitive data or are externally exposed, and mapping threats and protections across Service-Based Architecture and non-Service-Based Architecture connectivity. It discusses defenses such as Transport Layer Security, Internet Protocol Security, and Open Authorization for Network Function interactions, and highlights post-quantum considerations for key exchange and tunnel establishment.

Mehic et al. [10] provide a comprehensive overview of quantum cryptography for Fifth Generation networks, covering Quantum Key Distribution deployment options, integration challenges, and standardization aspects in cellular architectures. It contrasts Quantum Key Distribution with PQC and discusses how quantum-resilient key establishment can complement mechanisms such as TLS and Internet Protocol Security (IPsec) in Fifth Generation systems.

Lawo et al. [11] designs and implements a Post Quantum Cryptography network stack on a Data Processing Unit platform, combining Falcon with Kyber and Dilithium with Kyber for authentication and key establishment. It benchmarks performance on the NVIDIA Data Processing Unit platform, reporting latency and throughput trade-offs for PQC-protected communication.

Scalise et al. [4] provide a systematic survey of security considerations in Fifth- and Sixth-Generation systems, organizing challenges and trends across confidentiality, integrity, and availability objectives, and introducing a Zero Trust Architecture perspective. It highlights emerging directions, including Post-Quantum Cryptography for key exchange, as well as the security implications of Software-Defined Networking, Network Function Virtualization, and Multi-access Edge Computing deployments.

Mangla et al. [12] review Fifth-Generation security vulnerabilities in a future quantum computing environment and survey quantum-based solutions to mitigate these challenges, while motivating the transition toward Sixth Generation. It discusses Quantum Key Distribution and related quantum security techniques and argues that classical public key cryptography must be complemented by quantum-resilient alternatives for next-generation industrial deployments.

Dolente et al. [13] perform an experimental vulnerability assessment of open-source Fifth Generation Core Network Function implementations, focusing on externally exposed

Network Functions such as the Access and Mobility Management Function and the Network Repository Function with Network Exposure Function. It applies Application Programming Interface injection and fuzzing-style tests to Open5GS and OpenAirInterface, and reports robustness issues that can lead to denial-of-service attacks and other security risks.

Pell et al. [14] study service classification of Fifth Generation core network traffic using Machine Learning over flow and packet metadata, motivated by detecting malicious signaling without Deep Packet Inspection. It compares multiple Machine Learning models and shows how accurate service-level inference can support security monitoring for distributed Network Functions in Fifth-Generation core networks.

In Table I, 5G / SBA indicates an explicit focus on the 6G and Service-Based Interface (SBI) and TLS, IPsec, and Open Authorization (OAuth) indicate whether TLS, IPsec, and OAuth are discussed or used. 5G Stack indicates use of an open-source Fifth Generation implementation or testbed; Application Programming Interface (API)/(Fuzzing or Fuzz testing) - Automated Software Testing Technique (Fuzz) indicates API injection or fuzzing-style testing; and PQC indicates explicit PQC discussion or implementation.

Although current literature explores theoretical PQC and 5G vulnerabilities, practical non-disruptive deployments in operational 5G cores remain underexplored. To address this gap, the next section defines the threat model and the security goals that guide our architecture.

III. THREAT MODEL AND SECURITY GOALS

The threat model considers a passive adversary capable of recording encrypted traffic for later decryption in a HNLD scenario, as well as an active adversary performing Man-in-the-Middle (MITM) attacks to intercept, replay, or modify signaling data. We assume that the NFs and their host environments are trusted, while the underlying transport network is inherently insecure. While trusted NF execution is assumed, integration with Trusted Execution Environments (TEEs) or confidential-computing mechanisms represents an important direction for mitigating endpoint compromise in future deployments.

The protected assets are the confidentiality and integrity of SBI control-plane signaling exchanged between NFs (e.g., registration, discovery, and heartbeat) over potentially untrusted transport links. Consequently, the primary security objectives are to ensure long-term confidentiality, robust mutual authentication, and message integrity through post-quantum mechanisms. Furthermore, the architecture must support crypto-agility, enabling an incremental migration to PQC without modifying the internal logic of the NFs.

This model aligns with 3rd Generation Partnership Project (3GPP) assumptions where protection is applied at the SBA level; however, we reinforce these interfaces against quantum-capable adversaries. Out-of-scope threats include endpoint compromise, Denial of Service (DoS) attacks, side-channel analysis, and radio-access security.

By design, the sidecar introduces an additional trusted component in the NF pod, so reducing its attack surface (min-

TABLE I: Short State of the art survey.

Paper	5GC/SBA	TLS	IPsec	OAuth	Exp.	5G Stack	API/Fuzz	PQC
Mahyoub et al. [9]	●	●	●	●	○	○	○	●
Mehic et al. [10]	●	●	●	○	○	○	○	●
Lawo et al. [11]	○	○	○	○	○	○	○	●
Scalise et al. [4]	●	●	●	●	○	○	○	●
Mangla et al. [12]	○	●	●	○	○	○	○	●
Dolente et al. [13]	●	●	●	●	○	●	●	○
Pell et al. [14]	●	○	○	○	●	●	○	○
This work	●	●	●	●	●	●	●	●

imal codebase, least-privilege access, and hardened container configuration) is a key operational requirement.

IV. EMPOWERING THE CORE 5G WITH PQC

The transition to PQC is a key step in telecommunications security, but direct integration into the network ecosystem often requires complex modifications to legacy software, leading to compatibility issues and increased maintenance costs. To overcome these challenges, this work proposes a sidecar proxy pattern [15] that decouples security functions from the core business logic of the NF. Instead of modifying the NF source code, specialized proxies are deployed alongside each component to handle PQC-protected handshakes and key exchanges transparently. This modular layer abstracts the complexities of quantum-resistant protocols, creating a "plug-and-play" environment where algorithms can be updated without disrupting the underlying 5G infrastructure.

For this implementation, the ML-KEM-768 and ML-DSA-65 algorithms were selected due to their optimal balance between post-quantum resilience and operational efficiency, aligning with NIST Security Level 3 standards [16]. While higher-level parameters (e.g., Level 5) offer superior security margins, they introduce prohibitive communication overhead and handshake latency [17] that could compromise the strict performance requirements of the 5G SBA. Furthermore, these parameter sets benefit from mature implementations and extensive optimizations for digital signature calculations and verification [18]. By integrating these standardized algorithms via the sidecar approach, the architecture ensures robust, quantum-resistant protection while maintaining the high-performance throughput and low-latency response times essential for mission-critical core network operations.

While the architecture supports PQC encapsulation for N2 signaling, the experimental evaluation presented in this work focuses exclusively on SBI interactions to isolate control-plane cryptographic overhead.

V. THE PQC ENABLED FREE5GC

The experimental environment was built upon the free5GC platform, an open-source implementation of 3GPP 5G Core specifications designed for cloud-native deployments. Utilizing the free5gc-compose project, we implemented a modular post-quantum infrastructure using a sidecar proxy pattern that strictly separates cryptographic operations from the NF business logic. As illustrated in the architecture (see Figure 1),

the deployment relies on a specialized cryptolib container integrating liboqs v0.15.0 and the OpenSSL Open Quantum Safe (OQS)-provider, establishing the foundation for quantum-resistant primitives. To abstract these low-level interactions, we developed wrapper-kem and wrapper-sign services that expose REpresentational State Transfer Application Programming Interface (RESTful) APIs for ML-KEM and ML-DSA operations. This structure (see Figure 3), orchestrated via Docker Compose, ensures that cryptographic lifecycles, including the generation of entity-specific keys and certificates by a dedicated PQC-key-generator, are managed independently of the NF.

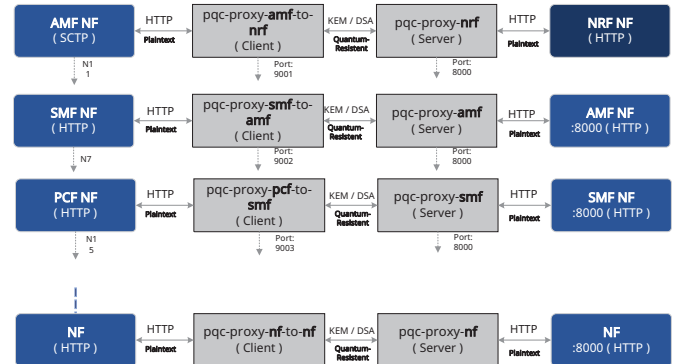


Fig. 1: PQC Secure Free5GC Architecture (partial view)

The core of the PQC integration lies in the pqc-proxy instances deployed in both server and client modes. NFs communicate via plain-text HTTP with local proxies within isolated Docker bridge networks (172.17.0.0/16), while inter-proxy traffic across network edges is encapsulated in ML-KEM-768 tunnels and authenticated via ML-DSA-65. A critical technical challenge addressed was the protection of the N2 interface; since NGAP signaling typically relies on Stream Control Transmission Protocol (SCTP), we deployed socat bridges to perform SCTP-to-Transmission Control Protocol (TCP) conversion, enabling PQC encapsulation between the pqc-proxy-gnb and pqc-proxy-amf-n2 (see Figure 2). To ensure traffic redirection, the free5GC configuration files (e.g., amfcfg.yaml, nrfcfg.yaml) were modified to route SBI requests through the local proxy's listener port.

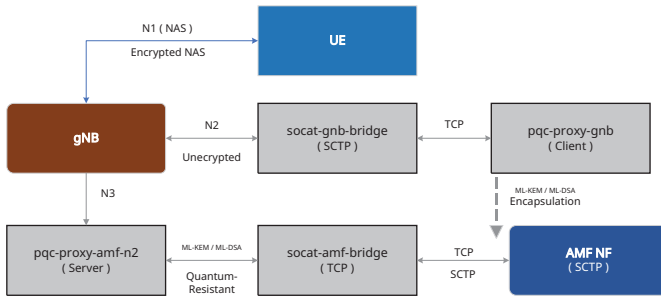


Fig. 2: PQC Radio Access Network (RAN) domain (partial view).

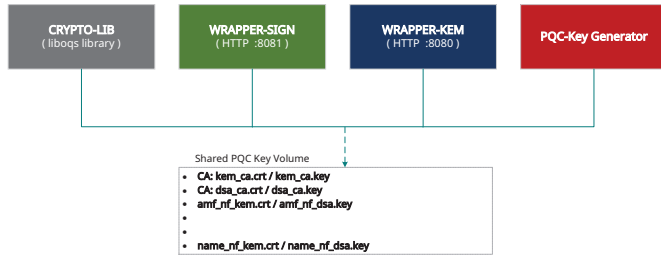


Fig. 3: Cryptographic infrastructure (partial view).

To evaluate the resulting cryptographic overhead and system stability, a robust monitoring framework was integrated directly into the architecture. This stack comprises Prometheus for metrics collection, Grafana for real-time visualization, Node Exporter for hardware telemetry, and cAdvisor for container-level resource tracking. This integrated observability framework enables precise, high-fidelity analysis of CPU and memory consumption, as well as latency variations across the PQC mechanisms deployed within the 5G Core, providing the necessary data to validate the performance trade-offs of the proposed sidecar approach (see Figure 4).

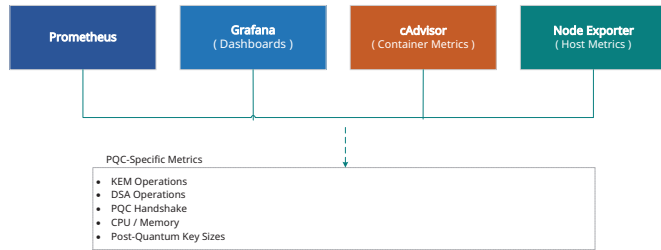


Fig. 4: Monitoring infrastructure layer.

VI. EXPERIMENTAL EVALUATION

This section presents the experimental methodology adopted to assess the performance implications of integrating PQC into the free5GC using the proposed sidecar architecture. The evaluation is designed to ensure reproducibility, isolate cryptographic overhead from wrapper-induced processing costs, and enable a controlled comparison between Native, classical TLS, and PQC-enabled deployments. The

following subsections describe the workload definition, testbed configuration, experimental design, and statistical reporting framework.

A. Evaluation Scope and Workload

To quantify the performance impact of the sidecar architecture and PQC, the evaluation focuses on control-plane SBI interactions between NFs and the Network Repository Function (NRF), including registration, discovery, and heartbeat procedures. By isolating the control plane from User Equipment (UE) emulation overhead, the setup enables direct comparison between the Native architecture, a classical TLS sidecar proxy, and a PQC sidecar proxy integrated via liboqs.

The workload consists of sequential HTTP/2 request-response exchanges per NF pair. Each experimental repetition executes 100 transactions without artificial concurrency to isolate cryptographic overhead from scheduling or load-induced effects. A transaction is considered successful when the complete exchange finishes without timeout or retransmission.

B. Testbed Configuration

All architectures were deployed in standardized Virtual Machines (VMs) running Ubuntu 22.04 Long Term Support (LTS), each allocated 4 CPU cores and 8 GB of Random Access Memory (RAM). The 5G Core was orchestrated using free5GC-compose v4.2.0. The PQC implementation used OpenSSL v3.4.0 integrated with liboqs v0.15.0, enabling ML-KEM-768 and ML-DSA via sidecar proxies. All experiments were conducted without hardware acceleration, isolating software-side overheads.

C. Experimental Design

For each architecture (Native, classical TLS Proxy, and PQC Proxy), experiments were repeated 30 times, each repetition involving a full container lifecycle reinitialization to capture variability due to virtualized resource allocation and network stack initialization. Within each repetition, 100 sequential SBI interactions were executed. The PQC configuration was evaluated across three CA parameter sets (ML-DSA-44, 65, and 87), while maintaining ML-KEM-768 and ML-DSA-65 for key exchange and signatures.

D. Metrics and Statistical Reporting

Metrics were collected using Prometheus, cAdvisor, and Node Exporter. For statistical reporting, each repetition is treated as an independent sample. Results are expressed using median and Interquartile Range (IQR) across repetitions, and 95% confidence intervals were computed over per-run median latency values using the *t*-distribution. The monitored performance and security indicators are summarized in Table II.

VII. RESULTS AND DISCUSSION

We evaluated three free5GC deployments: (i) *Native*, using standard Hypertext Transfer Protocol Security (HTTPS)/TLS; (ii) *TLS Proxy*, using a sidecar model with OpenSSL certificates; and (iii) *PQC Proxy*, enabling PQC through the proposed sidecar approach.

TABLE II: Selected metrics for evaluation.

Visual Analysis	Selected Metrics	Evaluation Goal
Latency Distribution	L_{SBI}	Assess delay scaling and jitter across architectures.
Resource Efficiency	CPU% Mem_{MB}	Validate hardware sustainability for PQC workloads.
Latency Breakdown	L_{Srv} L_{Cli}	Isolate sidecar overhead from cryptographic processing.
SBI Matrix	L_{SBI} per NF pair	Identify bottlenecks in 5G signaling (e.g., AMF, NRF).
PQC Operations	T_{enc} T_{sig} T_{ver}	Correlate KEM/DSA complexity with E2E performance.

Latency is measured at the HTTP/2 request–response boundary on the SBI: timestamps are captured immediately before request transmission and after complete response reception at the application layer. The SBI Communication metric, therefore, captures end-to-end control-plane transaction latency for one NF-to-NF interaction. Container Metrics capture proxy-side processing behavior, while Overall Statistics correspond to the per-run aggregate of all collected SBI request–response latencies across monitored NF pairs (Figure 5). Per-NF-pair heatmaps report mean latencies for individual paths (e.g., 5–6 ms in the Native case), whereas Figure 5 summarizes distributional medians aggregated across interactions within each run.

For statistical reporting, 95% confidence intervals were computed over per-run median latency values using the t -distribution across 30 independent repetitions, while error bars represent the IQR ($Q1$ – $Q3$) around the median. Native establishes the baseline, with SBI Communication and Container Metrics medians of 15.07 (Confidence Interval (CI): [15.0, 15.1] and [15.0, 15.2]) and minimal quartile deviation (± 0.05 – 0.1). Overall Statistics records 5.04 (CI: [4.9, 5.2]), indicating stable conventional TLS behavior.

The TLS Proxy converges SBI Communication and Container Metrics at 9.29 ms (CI: [9.3, 9.3] for both), corresponding to a 32.3% reduction relative to Native and near-zero IQR. This reduction does not imply lower cryptographic complexity; it is consistent with the sidecar centralizing connection management and reusing pooled HTTP/2/TLS sessions across repeated SBI exchanges, thereby amortizing handshake and transport setup costs. To ensure comparability, latency was measured at the same request–response boundary in all architectures, using identical keep-alive and connection-reuse settings across runs.

In contrast, the PQC Proxy increases SBI Communication to 54.05 ms (CI: [53.9, 54.1]) while preserving tight quartile bounds (± 0.1), quantifying a stable overhead of approximately 48–49 ms relative to classical TLS. Container Metrics remain stable at 9.46 ms (CI: [9.4, 9.5]), suggesting bounded resource behavior even under ML-KEM and ML-DSA processing (Figure 6).

From a standards perspective, the measured ~ 54 ms end-to-end SBI latency remains well within 5G control-plane timing tolerances, which are governed by NAS procedure timers and are typically on the order of seconds. Therefore,

the additional ~ 48 – 49 ms introduced by the PQC sidecar is operationally compatible with registration and authentication signaling, while still motivating optimization for latency-sensitive deployments.

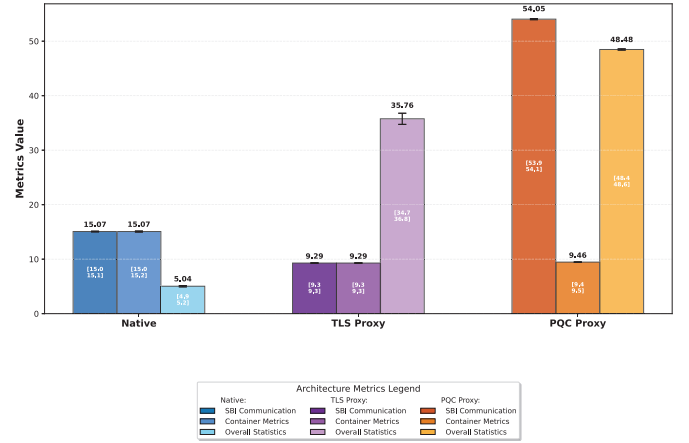


Fig. 5: Metrics comparison across architectures Core 5G.

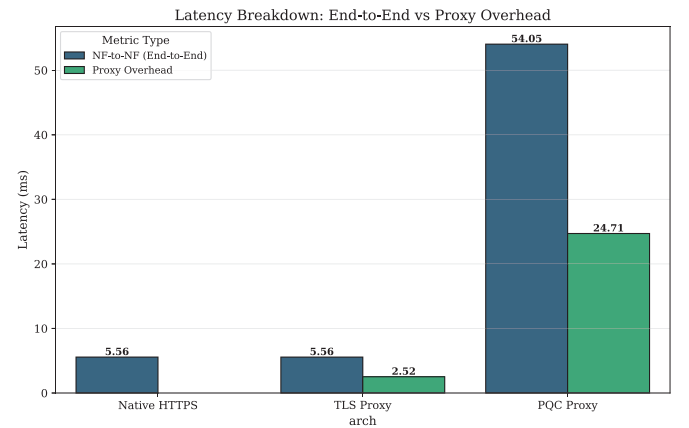


Fig. 6: Processing latency End-to-End.

Quartile analysis highlights distinct distributional behavior across architectures. Native exhibits low-variance distributions, while TLS Proxy shows compressed quartiles for SBI and container-level metrics but expanded quartiles for Overall Statistics, indicating aggregation variability in the proxy pipeline. By contrast, PQC Proxy maintains narrow quartile ranges across all metrics ($IQR \leq 0.2$), demonstrating higher absolute latency with bounded variance—a key property for predictable Quality of Service (QoS). The separation is clear: PQC SBI quartiles [53.9, 54.1] do not overlap with Native [15.0, 15.1] or TLS Proxy [10.2, 10.2], and the observed $CV < 0.4\%$ confirms stable behavior.

Heatmaps further corroborate these trends. Native maintains uniformly low SBI latencies (5.5–6.0 ms) across NF pairs (Figure 7), while TLS Proxy remains within the same order of magnitude (5.0–7.7 ms) and exhibits stable behavior consistent with optimized connection reuse (Figure 8). PQC Proxy exhibits a nearly uniform increase to 53.0–54.4 ms

across all NF pairs (Figure 9), indicating deterministic overhead largely independent of the communication pattern and dominated by ML-KEM-768 encapsulation/decapsulation and ML-DSA processing.

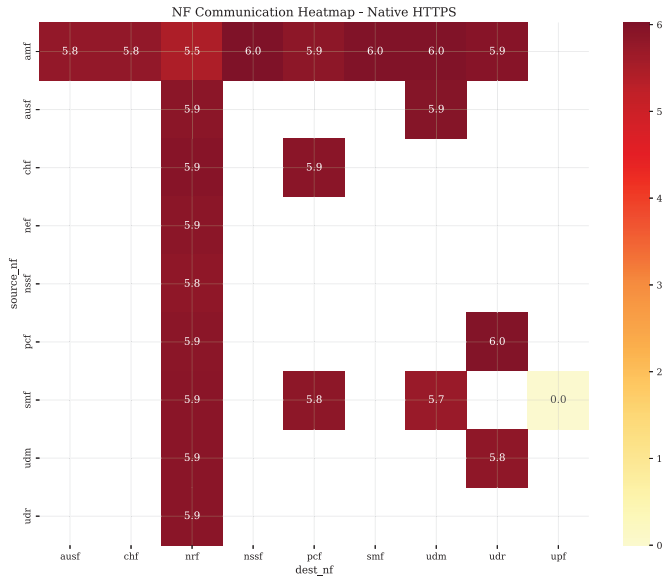


Fig. 7: Mean latency in the HTTPS baseline.

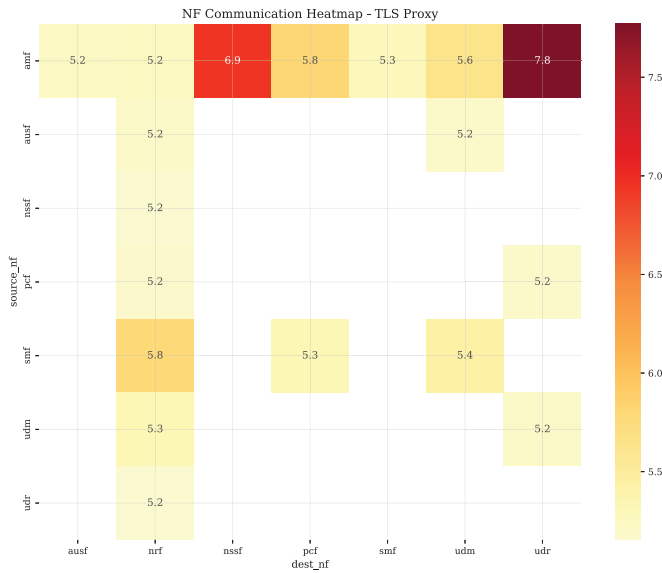


Fig. 8: Mean latency in the TLS Proxy.

Detailed latency breakdown for representative NRF-centric exchanges (Figures 10–12) shows the same pattern: Native remains near 5–6 ms, TLS Proxy remains near 5 ms with a small proxy processing component (2.43 ms), and PQC Proxy increases to 53–54 ms. The proxy processing estimate is stable (24.71 ms) across evaluated pairs, indicating deterministic overhead consistent with encapsulation/decapsulation dominance. The proxy processing estimate is computed inside the sidecar as $\Delta t = t_{\text{egress}} - t_{\text{ingress}}$, capturing cryptographic encapsulation/decapsulation and certificate validation time within

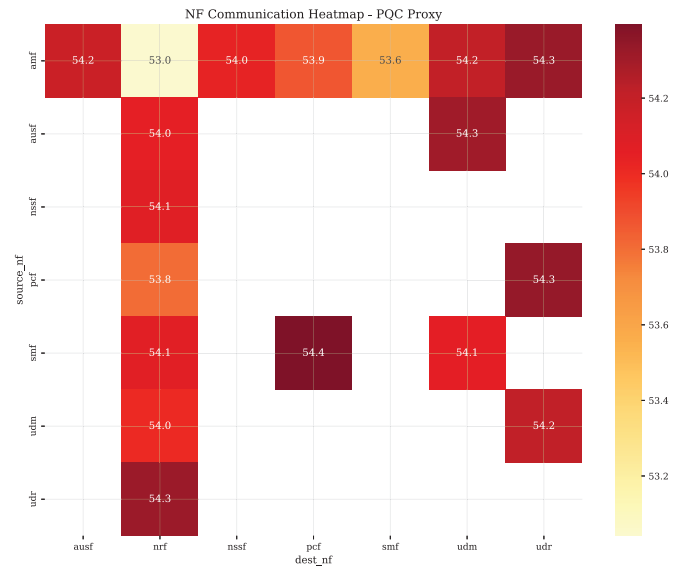


Fig. 9: Mean latency in the PQC Proxy

the proxy/wrappers, independent of NF application processing.

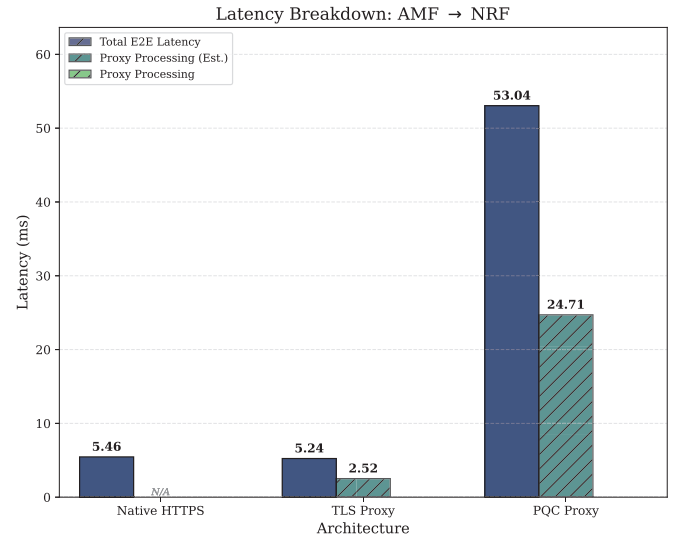


Fig. 10: Processing latency - AMF to NRF

The proxy processing breakdown (Figure 13) reveals asymmetric costs for TLS Proxy (server-side 3.99 ms vs. client-side 1.00 ms), consistent with server-dominated certificate validation and private-key operations. For the PQC Proxy, server-side (7.03 ms) and client-side (6.56 ms) costs become nearly symmetric (13.59 ms total), indicating that encapsulation and decapsulation impose comparable computational burdens and contribute directly to end-to-end latency. Overall PQC overhead comprises (i) ML-KEM encapsulation/decapsulation, (ii) ML-DSA signature generation/verification, and (iii) proxy-side certificate-chain validation.

Detailed analysis indicates that higher security levels in the CA (e.g., ML-DSA-87) directly increase server-side validation latency. To quantify the impact of security margins on opera-

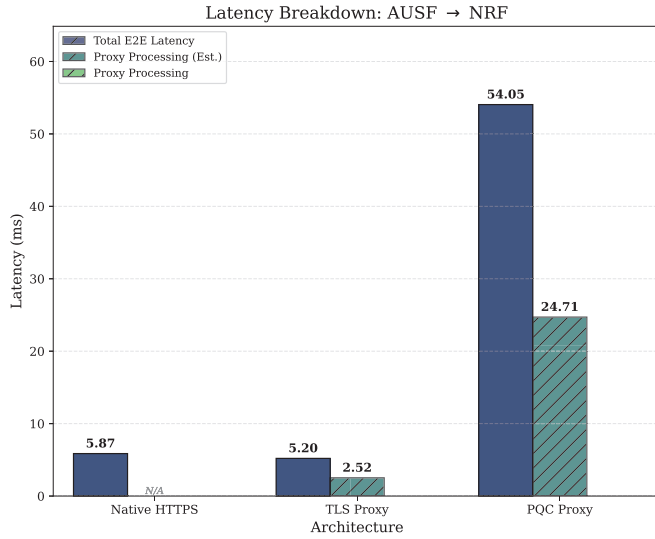


Fig. 11: Processing latency - Authentication Server Function (AUSF) to NRF

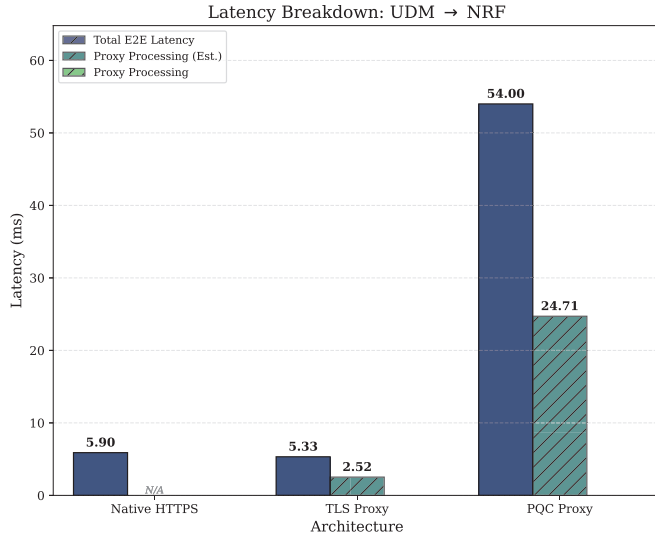


Fig. 12: Processing latency - UDM to NRF

tional latency, we evaluated three CA parameter levels within the PQC Proxy configuration.

As shown in Table III, increasing the CA security level yields a monotonic rise in server-side validation latency: the transition to ML-DSA-87 adds +1.40 ms to the median total SBI latency relative to ML-DSA-44. While this variation is smaller than the dominant PQC overhead, it is repeatable and measurable, confirming certificate-chain validation as a tunable contributor to end-to-end delay. Consequently, CA parameter selection serves as a concrete deployment lever for balancing post-quantum security margins and control-plane latency constraints.

We assume a trusted key-generation and distribution service; operational deployments should harden this trust anchor by using Hardware Security Modules (HSMs) and audited,

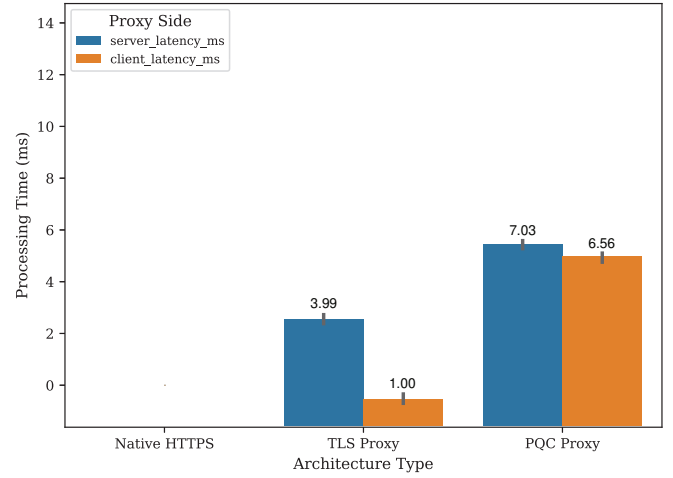


Fig. 13: PQC latency operations

TABLE III: Impact of CA security level on server-side validation latency (PQC Proxy).

CA Algorithm	Server Proxy (ms)	Total SBI Median (ms)	Δ vs ML-DSA-44 (ms)
ML-DSA-44	6.48	53.72	-
ML-DSA-65	7.03	54.05	+0.33
ML-DSA-87	8.21	55.12	+1.40

least-privilege key management to reduce the risk of a single point of compromise.

From a deployment perspective, the results indicate that PQC-sidecar integration remains feasible for control-plane operations with moderate latency tolerance, whereas authentication-intensive NF interactions may require resource scaling and/or optimized certificate-validation strategies. Lower CA security levels provide measurable latency reduction, highlighting a tunable trade-off between cryptographic strength and operational performance.

VIII. CONCLUSION

This work addressed the practical challenge of securing cloud-native 5G core signaling against future quantum threats, including harvest-now-decrypt-later and active MITM attacks, while preserving compatibility with existing NFs implementations. To bridge the gap between NIST PQC standardization and deployable telecommunications security, we proposed and implemented a sidecar proxy architecture that transparently provides post-quantum protection for SBI communication without modifying legacy NF codebases, enabling cryptographic agility through a modular security layer.

Using free5GC, we experimentally compared three scenarios, Native HTTPS, TLS Sidecar (Proxy/Wrapper), and PQC Sidecar (Proxy/Wrapper), under a controlled SBI workload centered on NF registration, discovery, and heartbeat exchanges. The results show that the PQC sidecar increases the end-to-end SBI latency to approximately 54 ms, corresponding to an additional 48–49 ms compared to classical TLS. However, the overhead remains deterministic and largely

uniform across NF pairs, with stable distributions that support predictable capacity planning. These findings indicate that transparent PQC insertion via sidecars is operationally viable for control plane signaling with moderate latency tolerance and provides a concrete, immediately applicable migration path that preserves operator investments in the current 5G infrastructure.

Our evaluation focused on control-plane traffic and isolated SBI interactions, leaving user-plane protection outside the current scope. The reliance on a trusted key generation and distribution service introduces a critical dependency that must be hardened for production deployments.

Future work should therefore extend PQC coverage to user-plane paths, evaluate hardware acceleration and certificate-validation optimizations for ML-KEM and ML-DSA, and investigate how performance differs if key sizes are chosen to roughly match specific security levels. Furthermore, future research should explore hybrid classical and post-quantum configurations during transition periods, and validate the approach under denser operational regimes, such as massive Internet of Things (IoT), while complementing the security analysis of the sidecar design with broader analysis of varied threat models.

Overall, by demonstrating a non-disruptive integration approach and quantifying its deterministic performance cost, this study provides a practical foundation for quantum-resistant 5G core deployments as the ecosystem advances toward 6G. In this sense, the proposed sidecar pattern translates NIST-standardized PQC into actionable deployment guidance for operators, enabling incremental adoption of quantum-resilient SBA signaling without invasive architectural changes.

ACKNOWLEDGMENT

The authors thank FAPEMIG (Grant #APQ00923-24), FAPESP MCTIC/CGI Research project 2018/23097-3 - SFI2 - Slicing Future Internet Infrastructures, Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001, PROPP/UFU and Fundação para a Ciência e Tecnologia (FCT) within the R&D Unit Project Scope UID/00319/2025 - Centro ALGORITMI (ALGORITMI/UM) for supporting this work.

REFERENCES

- [1] R. Zhou, H. Guo, F. E. C. Teo, and S. Bakiras, "A survey on post-quantum cryptography for 5g/6g communications," in *2023 IEEE International Conference on Service Operations and Logistics, and Informatics (SOLI)*, 2023, pp. 1–6.
- [2] S. Hoque, A. Aydeger, E. Zeydan, and M. Liyanage, "Analysis of post-quantum cryptography in user equipment in 5g and beyond," in *2025 IEEE 50th Conference on Local Computer Networks (LCN)*, 2025, pp. 1–9.
- [3] S. P. Sanon and H. D. Schotten, "Securing Mobile Networks in the Quantum Era: Imperative Role of Post-Quantum Cryptography," in *2025 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*, Jun. 2025, pp. 721–726, iSSN: 2575-4912. [Online]. Available: <https://ieeexplore.ieee.org/document/11037057>
- [4] P. Scalise, M. Boeding, M. Hempel, H. Sharif, J. Delloiacovo, and J. Reed, "A systematic survey on 5g and 6g security considerations, challenges, trends, and research areas," *Future Internet*, vol. 16, no. 3, 2024. [Online]. Available: <https://www.mdpi.com/1999-5903/16/3/67>

- [5] A. Olushola and S. P. Meenakshi, "Design and implementation of an authenticated post-quantum session protocol using ml-kem (kyber), ml-dsa (dilithium), and aes-256-gcm," *Frontiers in Physics*, vol. Volume 13 - 2025, 2026.
- [6] Q. D. Truong, H. Nguyen, T. Tan Nguyen, and H. Lee, "Nist post-quantum cryptography standards: A comprehensive review of theoretical foundations and implementations," *IEEE Access*, vol. 14, pp. 14 069–14 097, 2026.
- [7] C. Zong, "The mathematical foundation of post-quantum cryptography," *Research*, vol. 8, p. 0801, 2025. [Online]. Available: <https://spj.science.org/doi/abs/10.34133/research.0801>
- [8] A. Astarloa, J. Lázaro, and J. I. Gárate, "Crystals-dilithium post-quantum cyber-secure soc for wired communications in critical systems," *Internet of Things*, vol. 33, p. 101656, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2542660525001702>
- [9] M. Mahyoub, A. AbdulGhaffar, E. Alalade, E. Ndubisi, and A. Matrawy, "Security analysis of critical 5g interfaces," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 4, pp. 2382–2410, 2024.
- [10] M. Mehic, L. Michalek, E. Dervisevic, P. Burdiak, M. Plakalovic, J. Rozhon, N. Mahovac, F. Richter, E. Kaljic, F. Lauterbach, P. Njemcevic, A. Maric, M. Hamza, P. Fazio, and M. Voznak, "Quantum cryptography in 5g networks: A comprehensive overview," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 1, pp. 302–346, 2024.
- [11] D. C. Lawo, R. Frantz, A. C. Aguilera, X. A. I. Clemente, M. P. Podleš, J. L. Imaña, I. T. Monroy, and J. J. V. Olmos, "Falcon/kyber and dilithium/kyber network stack on nvidia's data processing unit platform," *IEEE Access*, vol. 12, pp. 38 048–38 056, 2024.
- [12] C. Mangla, S. Rani, N. M. Faseeh Qureshi, and A. Singh, "Mitigating 5g security challenges for next-gen industry using quantum computing," *Journal of King Saud University - Computer and Information Sciences*, vol. 35, no. 6, p. 101334, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1319157822002373>
- [13] F. Dolente, R. G. Garroppo, and M. Pagano, "A vulnerability assessment of open-source implementations of fifth-generation core network functions," *Future Internet*, vol. 16, no. 1, 2024. [Online]. Available: <https://www.mdpi.com/1999-5903/16/1/1>
- [14] R. Pell, M. Shojafar, D. Kosmanos, and S. Moschogiannis, "Service classification of network traffic in 5g core networks using machine learning," in *2023 IEEE International Conference on Edge Computing and Communications (EDGE)*, 2023, pp. 309–318.
- [15] S. Aldas and A. Babakian, "Cloud-Native Service Mesh Readiness for 5G and Beyond," *IEEE Access*, vol. 11, pp. 132 286–132 295, 2023. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/10327727>
- [16] G. Alagic, M. Bros, P. Ciadoux, D. Cooper, Q. Dang, T. Dang, J. Kelsey, J. Lichtinger, Y.-K. Liu, C. Miller *et al.*, *Status report on the fourth round of the nist post-quantum cryptography standardization process*. US Department of Commerce, National Institute of Standards and Technology, 2025.
- [17] J. A. Montenegro, R. Rios, and J. Bonilla, "Comparative analysis of post-quantum handshake performance in quic and tls protocols," *Computer Networks*, vol. 275, p. 111957, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1389128625009223>
- [18] I. Gorbenko, Y. Kachko, and Y. Derevianko, "Optimization of digital signature calculation and verification operations for the fips 205 standard. part 2," *Radiotekhnika*, no. 222, p. 7–21, Sep. 2025. [Online]. Available: <http://rt.nure.ua/article/view/343463>